Status	Finished
Started	Friday, 16 October 2026, 3:48 PM
Completed	Friday, 16 October 2026, 3:50 PM
Duration	2 mins 13 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It does not work with negative numbers
c. It requires the list to be sorted ✓
d. It can only be applied to large lists

Question **2**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n ✓
c. n/2
d. log n

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By starting from the first element
c. By dividing the list length by two ✓
d. By comparing each element sequentially

Question **4**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Divide the list into two equal parts
- b. Compare the target element with the first element in the list
- c. Compare the target element with the middle element in the list ✓
- d. Sort the list

Question **5**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. An error is raised
- b. The first element is returned
- c. The search is considered unsuccessful ✓
- d. The last element is returned

Question **6**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n-1
- b. n ✓
- c. log n
- d. n/2

Question **7**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the search algorithm finishes
- b. When the element is found in the list ✓
- c. When the list contains duplicate elements
- d. When the list is sorted

Question **8**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Searching ✓
- b. Finding
- c. Discovering
- d. Mining

Question **9**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is large and sorted
- b. When the list is small and unsorted ✓
- c. When the list is large and unsorted
- d. When the list is small and sorted

Question **10**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 99 ✓
- b. 90 and 100
- c. 89 and 94
- d. 94 and 99

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The list is sorted
- b. The search continues in the left or right sublist ✓
- c. The search stops
- d. The search continues from the beginning

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Skipping every second element
c. Checking each element sequentially ✓
d. Sorting the list before searching

Question **13**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is somewhere in the middle of the array ✓
b. When the item is not the array at all
c. When the item is the last element in the array
d. Item is the last element in the array or item is not there at all

Question **14**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Hash search
c. Binary search ✓
d. Linear search

Question **15**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By comparing each element in the list sequentially ✓
c. By dividing the list into halves
d. By sorting the list first